Project repository.

1 Problem statement

The goal of this work is to price a path-dependent derivative by Monte-Carlo simulation of the underlying asset under the **Constant Elasticity of Variance (CEV)** model.

The CEV model generalizes geometric Brownian motion (GBM) by allowing the volatility to depend on the price level through an elasticity parameter γ. Under the risk-neutral measure the underlying asset S_t follows

$$dS_t = (r-q)\,S_t\,dt + \sigma\,S_t^{\gamma}\,dW_t, \tag{1}$$

where r is the risk-free rate, q the continuous dividend yield, σ the volatility scale, γ the elasticity of variance, and W_t a standard Brownian motion. For $\gamma = 1$ the model reduces to the Black–Scholes GBM; for $\gamma < 1$ volatility increases as the price falls, reproducing the leverage effect observed in equity markets.

2 Simulation scheme

The process is discretized on a uniform grid $0 = t_0 < t_1 < \cdots < t_m = T$ with step $\Delta t = T/m$ using the Euler–Maruyama scheme:

$$S_{t+1} = S_t + (r-q)\,S_t\,\Delta t + \sigma\,S_t^{\gamma}\,\sqrt{\Delta t}\,Z_t, \qquad Z_t \sim \mathcal{N}(0,1), \tag{2}$$

with the non-negativity constraint $S_{t+1} \leftarrow \max(S_{t+1}, 0)$ to keep simulated prices admissible. N independent paths are generated. Figure **??** shows a sample of simulated trajectories.

3 Pricing a cliquet option

A **cliquet** (ratchet) option pays the sum of capped/floored periodic returns over a set of reset dates. Here we consider a cliquet that pays, at each reset date $t_i = iT/n$ ($i = 1, \ldots, n$), the positive part of the price increment over the preceding period. Its discounted payoff along a path is

$$\text{Payoff} = \sum_{i=1}^{n} e^{-rt_i}\,\max(S_{t_i} - S_{t_{i-1}},\,0)\,. \tag{3}$$

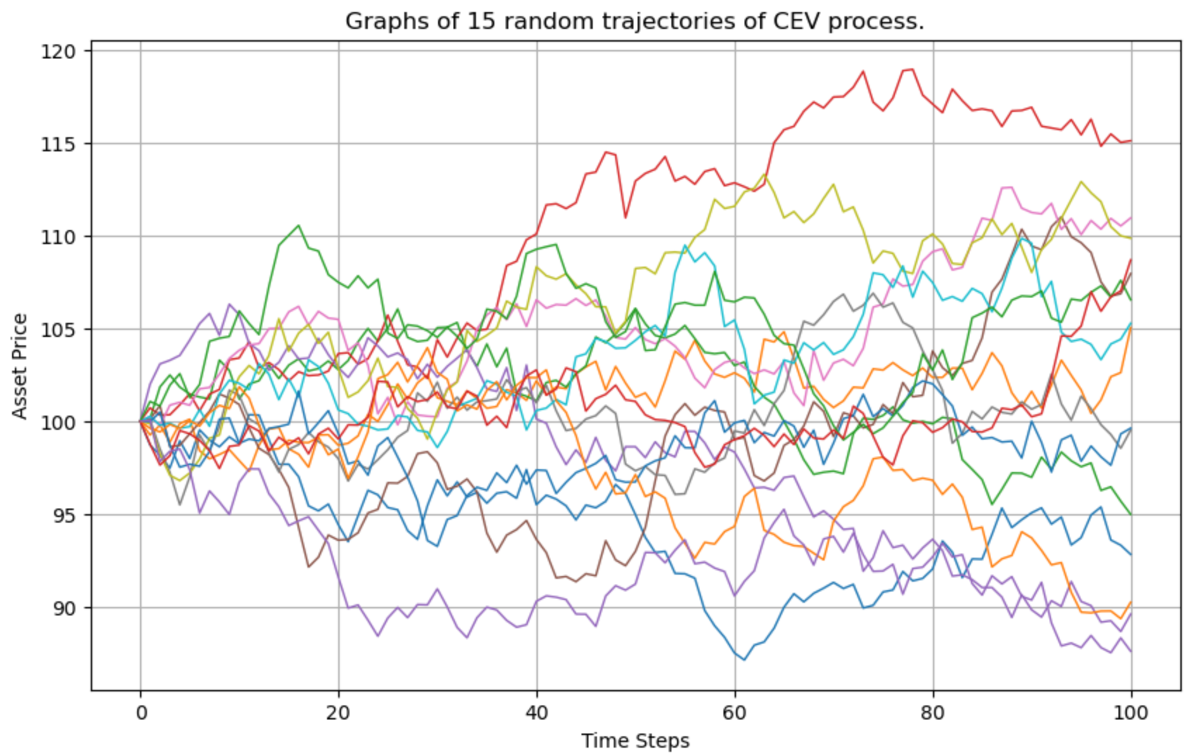

Figure 1: A sample of simulated CEV price trajectories.

The Monte-Carlo price estimate and its standard error are

$$\hat{P} = \frac{1}{N}\sum_{k=1}^{N} \mathrm{Payoff}^{(k)}, \qquad \widehat{\mathrm{SE}} = \frac{\hat{\sigma}_{\mathrm{Payoff}}}{\sqrt{N}}, \tag{4}$$

where $\hat{\sigma}_{\mathrm{Payoff}}$ is the sample standard deviation of the per-path payoffs. The standard error quantifies the Monte-Carlo sampling uncertainty and decreases as $O(N^{-1/2})$ with the number of paths.

4 Implementation

The code is organized as follows:

- `simulate_cev(S0, r, q, sigma, gamma, T, m, N)` — vectorized Euler–Maruyama simulation of N CEV paths on an m-step grid.
- `cliquet_price(...)` — Monte-Carlo estimate of the cliquet price and its standard error from the simulated paths.
- `plot_mc_paths(...)` — visualization of a subset of simulated trajectories.

The full experiments — parameter choices, price estimates, and convergence of the standard error with the number of paths — are in the accompanying notebook `main.ipynb`.

5 Conclusion

The work implements Monte-Carlo pricing of a path-dependent (cliquet) option under the CEV model, including a discretized simulation of the underlying, discounted payoff aggregation, and an estimate of the Monte-Carlo standard error. The framework is directly extensible to other path-dependent payoffs (Asian, barrier, lookback) by changing only the payoff function applied to the simulated paths.